

Decision under Uncertainty and Portfolio Analysis

Lecture 2: IRFA and MMMEF

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Source: Slides can be downloaded from [xiangyuqu@weebly.com](https://xiangyuqu.weebly.com).

The Allais choices

	\$5 million	\$1 million	\$0
a_1	0	1.00	0
a_2	0.10	0.89	0.01
a_3	0.10	0	0.90
a_4	0	0.11	0.89

Many people choose $a_1 \succ a_2$ but $a_3 \succ a_4$.

In-class question

Record both choices before proceeding. Which feature of the first pair is absent from the second?

Why the Allais pattern violates independence

Expected utility implies

$$\begin{aligned}a_1 \succ a_2 &\iff U(1) > .10U(5) + .89U(1) + .01U(0) \\ &\iff .11U(1) > .10U(5) + .01U(0),\end{aligned}$$

$$\begin{aligned}a_3 \succ a_4 &\iff .10U(5) + .90U(0) > .11U(1) + .89U(0) \\ &\iff .10U(5) + .01U(0) > .11U(1).\end{aligned}$$

The two final inequalities contradict one another.

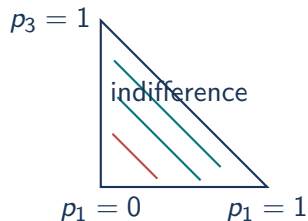
Behavioral reading

The move from certainty to a tiny chance of nothing receives special weight: the **certainty effect**.

The probability triangle

For three ordered outcomes, a lottery is a point in a triangle. Expected-utility indifference curves are straight and parallel because probabilities enter linearly.

- ▶ Parallel lines encode independence.
- ▶ Allais-type choices imply changing slopes.
- ▶ A common empirical pattern is **fanning out**.



Common Consequence Effect

Experiment 1

PROBLEM 1: Choose between

$$A : \begin{cases} 2,500 & \text{with probability } 0.33 \\ 2,400 & \text{with probability } 0.66 \\ 0 & \text{with probability } 0.01 \end{cases} \quad \text{and} \quad B : \begin{cases} 2400 & \text{with probability } 1. \end{cases}$$

PROBLEM 2: Choose between

$$C : \begin{cases} 2,500 & \text{with probability } 0.33 \\ 0 & \text{with probability } 0.67 \end{cases} \quad \text{and} \quad D : \begin{cases} 2400 & \text{with probability } 0.34 \\ 0 & \text{with probability } 0.66 \end{cases}$$

Common Consequence Effect

Experiment 1

PROBLEM 1: Choose between

$$A : \begin{cases} 2,500 & \text{with probability } 0.33 \\ 2,400 & \text{with probability } 0.66 \\ 0 & \text{with probability } 0.01 \end{cases} \quad \text{and} \quad B : \begin{cases} 2400 & \text{with probability } 1 \end{cases}$$

PROBLEM 2: Choose between

$$C : \begin{cases} 2,500 & \text{with probability } 0.33 \\ 0 & \text{with probability } 0.67 \end{cases} \quad \text{and} \quad D : \begin{cases} 2400 & \text{with probability } 0.34 \\ 0 & \text{with probability } 0.66 \end{cases}$$

Exercise

Demonstrate that the preferences $B \succ A$ and $C \succ D$ violates expected utility preferences.

Common Ratio Effect

Experiment 2

PROBLEM 1: Choose between

$$A : \begin{cases} 4,000 & \text{with probability 0.8} \\ 0 & \text{with probability 0.2} \end{cases} \quad \text{and} \quad B : \begin{cases} 3,000 & \text{with probability 1} \end{cases}$$

PROBLEM 2: Choose between

$$C : \begin{cases} 4,000 & \text{with probability 0.2} \\ 0 & \text{with probability 0.8} \end{cases} \quad \text{and} \quad D : \begin{cases} 3,000 & \text{with probability 0.25} \\ 0 & \text{with probability 0.75} \end{cases}$$

Exercise

Demonstrate that the preferences $B \succ A$ and $C \succ D$ violates independence axiom.

Isolation Effect

Problem 10 in KT

Consider the following two-stage game. In the first stage, there is a probability of 0.75 to end the game without winning anything, and a probability of 0.25 to move into the second stage. If you reach the second stage you have a choice between

(4,000; 0.80) and (3,000).

Your choice must be made before the game starts, i.e., before the outcome of the first stage is known.

Problem 4 in KT

Choose between

$C : \begin{cases} 4,000 & \text{with probability 0.2} \\ 0 & \text{with probability 0.8} \end{cases}$ and $D : \begin{cases} 3,000 & \text{with probability 0.25} \\ 0 & \text{with probability 0.75} \end{cases}$

Decision Trees for the Isolation Effect

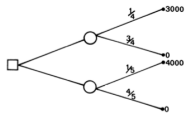


FIGURE 1.—The representation of Problem 4 as a decision tree (standard formulation).

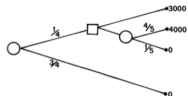


FIGURE 2.—The representation of Problem 10 as a decision tree (sequential formulation).

Probabilistic Insurance

Problem 9 in KT

Suppose you consider the possibility of insuring some property against damage, e.g., fire or theft. After examining the risks and the premium you find that you have no clear preference between the options of purchasing insurance or leaving the property uninsured. It is then called to your attention that the insurance company offers a new program called **probabilistic insurance**.

In this program you pay half of the regular premium. In case of damage, there is a 50 percent chance that you pay the other half of the premium and the insurance company covers all the losses; and there is a 50 percent chance that you get back your insurance payment and suffer all the losses. Recall that the premium for full coverage is such that you find this insurance barely worth its cost. Under these circumstances, would you purchase probabilistic insurance?

Another Description of Probabilistic Insurance

Notation: w wealth, L potential loss, π prob of loss, y insurance premium.

Indifference between accepting or rejecting initial policy:

$$\begin{pmatrix} w - L & \pi \\ w & 1 - \pi \end{pmatrix} \sim (w - y)$$

Probabilistic insurance generates the following lottery:

$$\begin{pmatrix} w - L & \pi/2 \\ w - y & \pi/2 \\ w - y/2 & 1 - \pi \end{pmatrix}$$

Would you choose the probabilistic insurance?

Another Description of Probabilistic Insurance

Notation: w wealth, L potential loss, π prob of loss, y insurance premium.

Indifference between accepting or rejecting initial policy:

$$\begin{pmatrix} w - L & \pi \\ w & 1 - \pi \end{pmatrix} \sim (w - y)$$

Probabilistic insurance generates the following lottery:

$$\begin{pmatrix} w - L & \pi/2 \\ w - y & \pi/2 \\ w - y/2 & 1 - \pi \end{pmatrix}$$

Would you choose the probabilistic insurance?

Majority (80% of subjects answered NO!)

Risk-Averse EU Agents Prefer Probabilistic Insurance

Notation: w wealth, L potential loss, π prob of loss, y insurance premium.

Indifference between accepting or rejecting initial policy:

$$\begin{pmatrix} w - L & \pi \\ w & 1 - \pi \end{pmatrix} \sim (w - y)$$

Mixing the two, independence implies

$$\begin{pmatrix} w - L & \pi/2 \\ w - y & \pi/2 \\ w & (1 - \pi)/2 \\ w - y & (1 - \pi)/2 \end{pmatrix} \sim (w - y)$$

Risk aversion (SOSD monotonicity) then implies

$$\begin{pmatrix} w - L & \pi/2 \\ w - y & \pi/2 \\ w - y/2 & 1 - \pi \end{pmatrix} \succeq (w - y)$$

Why Does Probabilistic Insurance SOSD the Mixture?

Let M be the mixture on the previous slide and P the probabilistic insurance lottery. Both retain probability $\pi/2$ on $w - L$ and $\pi/2$ on $w - y$. Only the remaining probability $1 - \pi$ changes:

$$\underbrace{\begin{cases} w - y & \text{with conditional probability } 1/2, \\ w & \text{with conditional probability } 1/2 \end{cases}}_{\text{remaining branch of } M} \longrightarrow \underbrace{w - y/2}_{\text{remaining branch of } P}.$$

A mean-preserving contraction

The risky branch is replaced by its mean: $\frac{1}{2}(w - y) + \frac{1}{2}w = w - y/2$. Thus $\mathbb{E}[P] = \mathbb{E}[M]$, while risk is reduced, so $P \succeq_{\text{SOSD}} M$.

For every increasing, concave u , the common outcomes cancel and Jensen's inequality gives

$$\mathbb{E}[u(P)] - \mathbb{E}[u(M)] = (1 - \pi) \left[u\left(w - \frac{y}{2}\right) - \frac{u(w - y) + u(w)}{2} \right] \geq 0.$$

Since $M \sim (w - y)$, we obtain $P \succeq (w - y)$. The preference is strict if u is strictly concave, $y > 0$, and $\pi < 1$.

Probabilistic Insurance versus Risk of Nonpayment

Risk of nonpayment: α chance of nonpayment with premium y' :

$$\begin{pmatrix} w - L - y' & \alpha\pi \\ w - y' & (1 - \alpha)\pi \\ w - y' & 1 - \pi \end{pmatrix}$$

- Evidence: people tend to require a much more than proportional decrease in premium to tolerate risk of nonpayment: $y' < (1 - \alpha)y$.
- Probabilistic insurance is similar to risk of nonpayment with $\alpha = 0.5$ and $y' = y/2$ (but the two are not quite the same!):

$$\begin{pmatrix} w - L & \pi/2 \\ w - y & \pi/2 \\ w - y/2 & 1 - \pi \end{pmatrix} \quad \text{versus} \quad \begin{pmatrix} w - L - y/2 & \pi/2 \\ w - y/2 & \pi/2 \\ w - y/2 & 1 - \pi \end{pmatrix}$$

- A disproportional change in willingness to pay in response to nonpayment risk is difficult to match with EU, but does not yield a direct contradiction of EU like probabilistic insurance.

Machina's Generalization of the Common Consequence Effect

Machina (1987) describes a generalization of the CCE. Let δ_x be the Dirac measure that gives x with certainty, let P involve outcomes both greater and less than x , and let $Q \geq_{\text{FOSD}} R$.

Problem 1: Choose between

$$A : \alpha P + (1 - \alpha)Q \quad B : \alpha\delta_x + (1 - \alpha)Q.$$

Problem 2: Choose between

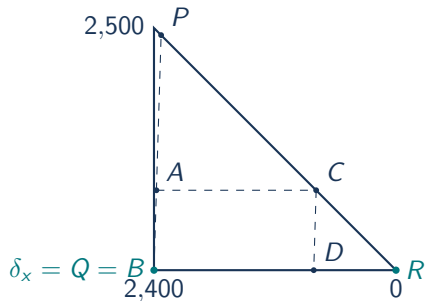
$$C : \alpha P + (1 - \alpha)R \quad D : \alpha\delta_x + (1 - \alpha)R.$$

- ▶ Independence requires $P \succ \delta_x \iff A \succ B \iff C \succ D$.
- ▶ Machina (1987) suggests violations tend to take the form $B \succ A$ and $C \succ D$.
- ▶ However, the evidence is actually mixed.

Illustration: Kahneman and Tversky (1979)

$$\begin{array}{ll} A : \alpha P + (1 - \alpha)Q & B : \alpha\delta_x + (1 - \alpha)Q \\ C : \alpha P + (1 - \alpha)R & D : \alpha\delta_x + (1 - \alpha)R \end{array}$$

The example has $x = 2,400$, $Q = \delta_x$, $R = \delta_0$, $\alpha = .34$, and $P = \frac{33}{34}\delta_{2,500} + \frac{1}{34}\delta_0$.



Machina's Hypothesis II

$$\begin{array}{ll} A : \alpha P + (1 - \alpha)Q & B : \alpha\delta_x + (1 - \alpha)Q \\ C : \alpha P + (1 - \alpha)R & D : \alpha\delta_x + (1 - \alpha)R \end{array}$$

- Independence requires $A \succeq B \iff C \succeq D$.
- What weaker restriction permits reversals $B \succ A$ and $C \succ D$, but rules out $A \succ B$ and $D \succ C$?

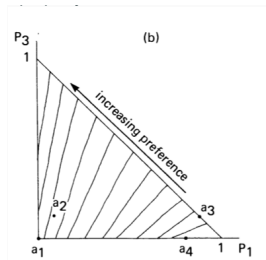
Machina (1982): Hypothesis II

If $Q \succeq_{\text{FOSD}} R$, then for any P and x ,

$$\begin{aligned} \alpha P + (1 - \alpha)Q &\succeq \alpha\delta_x + (1 - \alpha)Q \\ \implies \alpha P + (1 - \alpha)R &\succeq \alpha\delta_x + (1 - \alpha)R. \end{aligned}$$

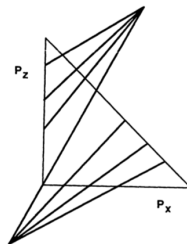
Hypothesis II and “Fanning Out” of Indifference Curves

Hypothesis II implies a “fanning out” property of indifference curves.



Machina (1982)

Others argue preferences exhibit a “mixed fan”: Hypothesis II holds for $P \succ Q$, but the pattern reverses for $Q \succ P$.



Gul (1991)

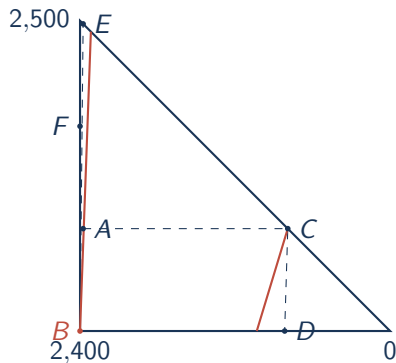
“Fanning Out” versus “Mixed Fan” versus Something Else?

Allais paradoxes concern the lower-right section of the simplex. There is less evidence about the upper-left section, which distinguishes fanning out from a mixed fan. An example of a test uses the new lotteries E and F :

Lottery	2,500	2,400	0
A	.33	.66	.01
B	0	1	0
C	.33	0	.67
D	0	.34	.66
E	.99	0	.01
F	.66	.34	0

- ▶ Choosing B, C, F is consistent with both fanning out and a mixed fan.
- ▶ Choosing B, C, E is consistent with a mixed fan, but not fanning out.

Illustration



Only $F \succ E$ is consistent with fanning out; either ranking is consistent with a mixed fan.

Second-Order Risk Aversion

An expected-utility maximizer with differentiable u is willing to bear at least a small amount of actuarially favorable risk. This is difficult to reconcile with:

- ▶ Avoiding some actuarially favorable investments.
- ▶ Purchasing full insurance coverage at actuarially unfair premiums.

Second-order risk aversion

For small zero-mean gambles, the risk premium is proportional to the variance σ^2 .

The patterns above can be explained by non-expected-utility models with **first-order risk aversion**: the risk premium of a small zero-mean gamble is proportional to its standard deviation σ .

Arrow–Pratt Approximation

To illustrate second-order risk aversion and its relationship to expected utility, we develop the Arrow–Pratt approximation.

- ▶ Initial wealth w .
- ▶ Risk $\tilde{x}$ with mean μ and variance σ^2 .
- ▶ Investment level (scale of risk) t .

Let $f(t)$ denote the certainty equivalent of random wealth $w + t\tilde{x}$:

$$u(f(t)) = \mathbb{E}[u(w + t\tilde{x})].$$

A second-order Taylor expansion of f around $t = 0$ gives the **Arrow–Pratt approximation** of the certainty equivalent.

Arrow–Pratt Approximation: Derivation

First, $u(f(t)) = \mathbb{E}[u(w + t\tilde{x})]$ implies $f(0) = w$. Differentiating with respect to t gives

$$\begin{aligned}u'(f(t))f'(t) &= \mathbb{E}[u'(w + t\tilde{x})\tilde{x}], \\u'(w)f'(0) &= u'(w)\mathbb{E}[\tilde{x}], \\f'(0) &= \mu.\end{aligned}$$

Differentiating again gives

$$\begin{aligned}u''(f(t))(f'(t))^2 + u'(f(t))f''(t) &= \mathbb{E}[u''(w + t\tilde{x})\tilde{x}^2], \\u''(w)\mu^2 + u'(w)f''(0) &= u''(w)\mathbb{E}[\tilde{x}^2], \\f''(0) &= \frac{u''(w)}{u'(w)}(\mathbb{E}[\tilde{x}^2] - \mu^2) = -A(w)\sigma^2,\end{aligned}$$

where $A(w) = -u''(w)/u'(w)$ is the Arrow–Pratt coefficient of absolute risk aversion.

Arrow–Pratt Approximation and Second-Order Risk Aversion

The second-order Taylor expansion around $t = 0$ is

$$\begin{aligned} f(t) &\approx f(0) + tf'(0) + \frac{1}{2}t^2f''(0) \\ &= w + t\mu - \frac{1}{2}t^2\sigma^2A(w). \end{aligned} \tag{1}$$

- ▶ $t\mu$ is the expected value of $t\tilde{x}$.
- ▶ $t\sigma$ is its standard deviation (for $t \geq 0$).

The mean has a first-order effect on the certainty equivalent, whereas the standard deviation has only a second-order effect.

Second-order risk aversion

If $\mu > 0$, some positive investment is preferable to $t = 0$, however large the finite variance σ^2 and coefficient $A(w)$ are.

Rabin (2000) Paradox

If a person rejects a 50–50 lose \$100/gain g bet at *all wealth levels*, then they also reject a 50–50 lose L /gain G bet with G as below.

Loss L	Small-bet gain g			
	\$101	\$105	\$110	\$125
\$400	400	420	550	1,250
\$600	600	730	990	∞
\$800	800	1,050	2,090	∞
\$1,000	1,010	1,570	∞	∞
\$2,000	2,320	∞	∞	∞
\$4,000	5,750	∞	∞	∞
\$6,000	11,810	∞	∞	∞
\$8,000	34,940	∞	∞	∞
\$10,000	∞	∞	∞	∞
\$20,000	∞	∞	∞	∞

- ▶ People tend to reject the small gambles, especially $g = 101$ or 105.
- ▶ Rejecting the implied large-scale gambles is unrealistic.

Key to the Result: Reject at All Wealth Levels

- ▶ Suppose a risk-averse individual rejects a $-\$100/ + \110 gamble at wealth $w = \$290,000$, and we know nothing else.
- ▶ Utility could be very concave around current wealth (perhaps kinked), but approximately linear elsewhere.
- ▶ Such an individual rejects larger proportional copies of the gamble, but might accept a proportionally larger gain, such as $-\$4,000/ + \$5,000$.
- ▶ The key to Rabin's conclusion is the stronger assumption that the individual rejects the $-\$100/ + \110 bet at **every wealth level**.
- ▶ Marginal utility then falls rapidly not just between $\$289,900$ and $\$290,110$, but between $w - \$100$ and $w + \$110$ for every w . The cumulative effect is very large!

A Rabin(-ish) Calibration Result

Suppose an individual has expected-utility preferences with an increasing, concave utility function u , and rejects a 50–50 gamble with loss l and gain g at every wealth level. Let $g > l > 0$ and m, n be positive integers.

1. If $(g/l)^m < 2$, she rejects a 50–50 gamble with loss $L = mg$ and gain $G = ng$ whenever

$$n < f(l, g, m) \equiv \frac{\log(2 - (g/l)^m)}{\log(l/g)}.$$

2. If $(g/l)^m > 2$, she rejects the gamble with loss $L = mg$ regardless of the possible gain.

Example: $l = 100$, $g = 105$

$f(100, 105, 9) = 16.43$: reject loss $L = 9(105) = 945$ and gain $G = 16(105) = 1,680$.

$(105/100)^{15} = 2.08 > 2$: reject any gamble with loss $L > 15(105) = 1,575$.

Proof I: The Key Inequality

Rejection of the small gambles implies

$$\begin{aligned} u(w) &\geq \frac{1}{2}u(w-l) + \frac{1}{2}u(w+g) \\ \iff u(w) - u(w-l) &\geq u(w+g) - u(w). \end{aligned}$$

By concavity,

$$u(w) - u(w-g) \geq \frac{g}{l} [u(w) - u(w-l)].$$

Together these yield the key equation for calibrating the decline of marginal utility:

$$u(w) - u(w-g) \geq \frac{g}{l} [u(w+g) - u(w)]. \quad (2)$$

Proof II: Bounding the Utility Loss

Since (2) holds at every wealth level,

$$\begin{aligned}u(w) - u(w - g) &\geq \frac{g}{l} [u(w + g) - u(w)], \\u(w - g) - u(w - 2g) &\geq \left(\frac{g}{l}\right)^2 [u(w + g) - u(w)], \\&\vdots \\u(w - (m - 1)g) - u(w - mg) &\geq \left(\frac{g}{l}\right)^m [u(w + g) - u(w)].\end{aligned}$$

Thus, for $L = mg$,

$$\begin{aligned}u(w) - u(w - L) &= \sum_{i=1}^m [u(w - (i - 1)g) - u(w - ig)] \\&\geq \sum_{i=1}^m \left(\frac{g}{l}\right)^i [u(w + g) - u(w)].\end{aligned}$$

Proof III: Bounding the Utility Gain

Using (2) in the other direction,

$$\begin{aligned}u(w + 2g) - u(w + g) &\leq \frac{l}{g}[u(w + g) - u(w)], \\u(w + 3g) - u(w + 2g) &\leq \left(\frac{l}{g}\right)^2 [u(w + g) - u(w)], \\&\vdots \\u(w + ng) - u(w + (n-1)g) &\leq \left(\frac{l}{g}\right)^{n-1} [u(w + g) - u(w)].\end{aligned}$$

Thus, for $G = ng$,

$$\begin{aligned}u(w + G) - u(w) &= \sum_{i=1}^n [u(w + ig) - u(w + (i-1)g)] \\&\leq \sum_{i=1}^n \left(\frac{l}{g}\right)^{i-1} [u(w + g) - u(w)].\end{aligned}$$

Proof IV: Calibration

$$u(w) - u(w - L) \geq \sum_{i=1}^m \left(\frac{g}{l}\right)^i [u(w + g) - u(w)],$$
$$u(w + G) - u(w) \leq \sum_{i=1}^n \left(\frac{l}{g}\right)^{i-1} [u(w + g) - u(w)].$$

Therefore, reject the 50–50 loss $L = mg$ /gain $G = ng$ gamble whenever

$$\sum_{i=1}^n \left(\frac{l}{g}\right)^{i-1} < \sum_{i=1}^m \left(\frac{g}{l}\right)^i$$
$$\iff \frac{1 - (l/g)^n}{1 - l/g} < \frac{(g/l)^{m+1} - g/l}{g/l - 1} = \frac{(g/l)^m - 1}{1 - l/g}$$
$$\iff (l/g)^n > 2 - (g/l)^m.$$

If $(g/l)^m < 2$, taking logs gives $n < \frac{\log(2 - (g/l)^m)}{\log(l/g)}$.

Rabin Paradox: Accounting for DARA

Repeat the calibration at initial wealth \$290,000, assuming rejection of the small bet only for $w \leq \$300,000$.

Loss L	Small-bet gain g			
	\$101	\$105	\$110	\$125
\$400	400	420	550	1,250
\$600	600	730	990	36,000,000,000
\$800	800	1,050	2,090	90,000,000,000
\$1,000	1,010	1,570	718,190	160,000,000,000
\$2,000	2,320	69,930	12,210,880	850,000,000,000
\$4,000	5,750	635,670	60,528,930	9,400,000,000,000
\$6,000	11,510	1,557,360	180,000,000	89,000,000,000,000
\$8,000	19,290	3,058,540	510,000,000	830,000,000,000,000
\$10,000	27,780	5,503,790	1,300,000,000	7,700,000,000,000,000
\$20,000	85,750	71,799,110	160,000,000,000	540,000,000,000,000,000

- With decreasing absolute risk aversion, small bets may become attractive at very high wealth levels. Recall: CRRA implies DARA.
- The paradox is robust to DARA, as the table shows.

Risk Aversion in Different Domains

The equity premium puzzle and Rabin's paradox relate risk aversion across different domains. CRRA expected utility illustrates the issue:

$$\mathbb{E}_t \left[\frac{c_{t+1}^{1-\alpha}}{1-\alpha} \right].$$

Domain	Scale of risk	Implied CRRA
First-order risk aversion	Infinitesimal	Infinite ($\alpha = \infty$)
Small Rabin gambles	Small	Huge ($\alpha \geq 45$)
Stock market risk	Moderate	Large ($\alpha \approx 18$)
Large Rabin gambles; occupational earnings risk; entrepreneurial risk	Large	Small ($\alpha \leq 10$)

CRRA utility is used for illustration, but the puzzles arise for any expected-utility preference.

References I

- ▶ Epstein, L. G., and S. E. Zin (1989). "Substitution, Risk Aversion, and the Temporal Behavior of Consumption and Asset Returns: A Theoretical Framework." *Econometrica*, 57, 937–969.
- ▶ Gul, F. (1991). "A Theory of Disappointment Aversion." *Econometrica*, 59, 667–686.
- ▶ Kahneman, D., and A. Tversky (1979). "Prospect Theory: An Analysis of Decision Under Risk." *Econometrica*, 47, 263–292.
- ▶ Kocherlakota, N. R. (1996). "The Equity Premium: It's Still a Puzzle." *Journal of Economic Literature*, 34, 42–71.
- ▶ Kreps, D. M., and E. L. Porteus (1978). "Temporal Resolution of Uncertainty and Dynamic Choice Theory." *Econometrica*, 46, 185–200.
- ▶ Machina, M. J. (1982). "'Expected Utility' Analysis without the Independence Axiom." *Econometrica*, 50, 277–323.

References II

- ▶ Machina, M. J. (1987). "Choice Under Uncertainty: Problems Solved and Unsolved." *Economic Perspectives*, 1, 121–154.
- ▶ Mehra, R., and E. Prescott (1985). "The Equity Premium: A Puzzle." *Journal of Monetary Economics*, 15, 145–161.
- ▶ Rabin, M. (2000). "Risk Aversion and Expected-Utility Theory: A Calibration Theorem." *Econometrica*, 68, 1281–1292.
- ▶ Weil, P. (1989). "The Equity Premium Puzzle and the Risk-Free Rate Puzzle." *Journal of Monetary Economics*, 24, 401–421.